

# HAT-MedMNIST

A hybrid agentic training framework for MedMNIST

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Technical Responsible · Project Manager · Reviewer

Agentic AI / medical-image benchmarks

- 1 **Motivation, Objectives & Plan**

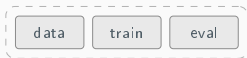
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- 2 **System Architecture & MVP**

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- 3 **Prior Art, Governance & Validation**

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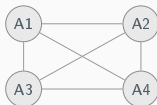
# Why HAT-MedMNIST

## Script pipeline



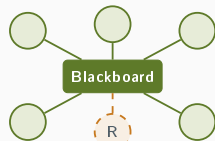
tightly coupled  
hard to audit

## Typical LLM agents



random at every step  
locked to one vendor

## HAT-MedMNIST



typed contracts · provenance  
independent reviewer with veto

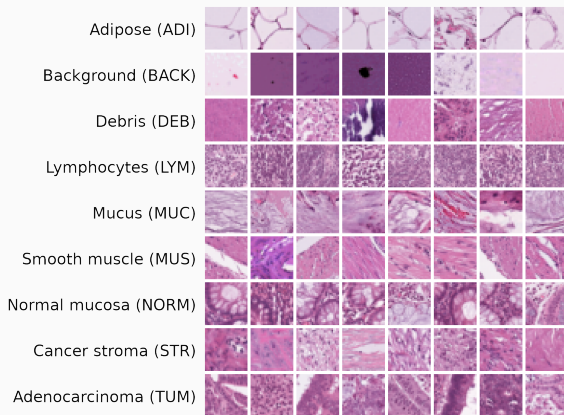
## The problem

- Training pipelines are monolithic scripts: hard to audit and to reproduce.
- LLM-agent frameworks let a model improvise every step, tied to one vendor.

## Our answer

- One duty per agent; agents exchange only versioned artefacts.
- LLM reasoning only at declared decision points, with a fallback.
- An independent reviewer audits every stage and can veto.

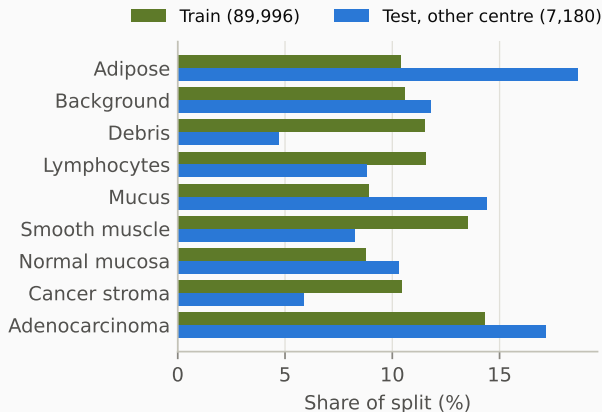
# The task in pictures: PathMNIST



Can software agents learn to tell colon tissue types apart?

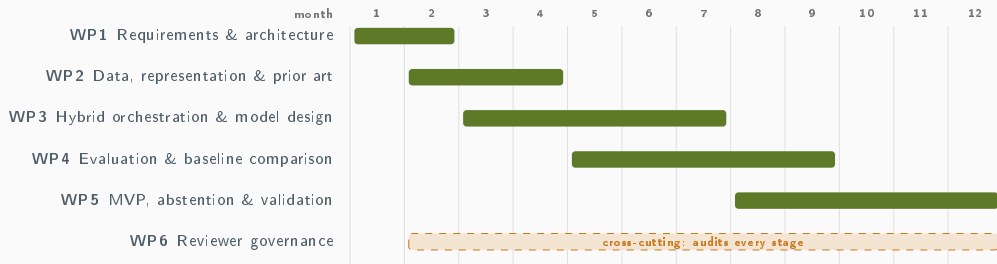
- PathMNIST: stained (H&E) patches from colorectal cancer slides, part of MedMNIST v2.
- 9 tissue classes, 107,180 images of only  **$28 \times 28$  pixels**.
- Train/validation from one source; test from **a different hospital**.
- Small images keep experiments fast; the second hospital keeps the test honest.

# The data at a glance



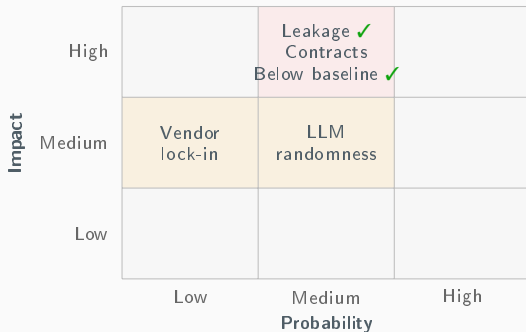
- Official split: 89,996 / 10,004 / 7,180 images.
- Training classes are near-balanced (largest/smallest = 1.63).
- The test hospital differs: adipose 18.6% vs 10.4%, debris 4.7% vs 11.5%.
- So test accuracy measures **transfer** to new data, not memory.

# The plan: six work packages in twelve months



- Hybrid Agile / stage-gate: short iterations, a formal gate at the end of each WP.
- Each WP delivers a reviewable output (D1–D6) with numeric KPIs.
- WP6, the reviewer, runs alongside every other WP.
- Goal: an integrated MVP demonstrated at **TRL 7** in month 12.

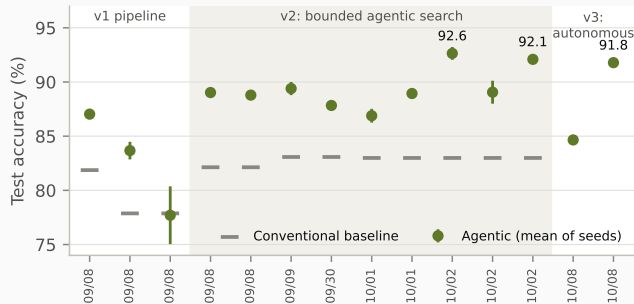
# Risk map



- **Leakage**: an audit gate blocks training until splits and statistics are checked.
- **Contracts**: typed artefacts fail fast; the orchestrator can veto.
- **Below baseline**: baseline-first, ablations, negative results reported.
- **LLM randomness**: temperature 0, logged provenance, fallback.
- **Lock-in**: any open model behind one interface.

✓ already demonstrated in the runs

# From first run to MVP: 14 runs in one month



- **v1** (8 Sep): first end-to-end runs; one tie with the baseline.
- **v2** (9 Sep–2 Oct): agents choose within a bounded search; every run beats the baseline, best 92.6%.
- **v3** (8 Oct): fully autonomous; agents write and launch their own experiment code.
- Same official split throughout, so the runs are comparable.

# The flagship run in numbers

**3 h 31 min**

end-to-end, full dataset

**8**

model trials, chosen by  
agents

**28**

LLM decisions, local model

**72**

versioned artefacts

## What ran

- 11 stages on the full dataset, one local CUDA machine.
- LLM: qwen3.8 served locally by Ollama; no cloud model.
- The search agent proposed and launched all 8 trials.

## Where the time went

- Model search: 3 h 20 min, 95% of the run.
- All other stages together: about 11 minutes.
- 56 recorded calls: 12 stage starts, 12 reviews, 32 worker jobs.

# Road to TRL 7



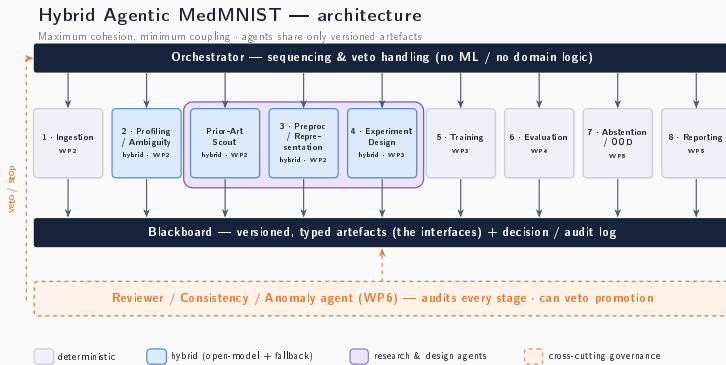
## Done

- Autonomous end-to-end run on the full dataset (8 Oct).
- 3 seeds, conventional baseline, ablations and live arXiv sources (2 Oct).

## Evidence still to file

- Fault-injection report on the current code.
- Replay of the flagship run; signed mitigation list for reviewer notes.
- Demonstration by someone other than the implementer.

# The Blackboard architecture



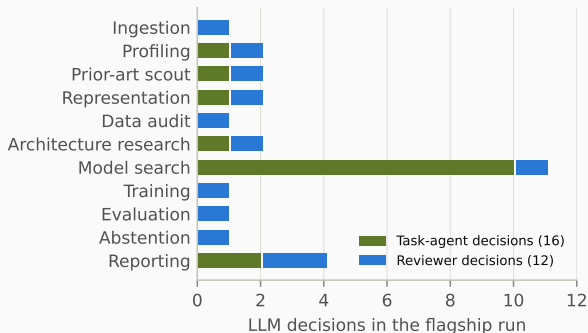
## Coordination

- **Orchestrator:** sequences stages; holds no domain logic.
- **Blackboard:** versioned, typed, checksummed artefacts plus a decision log.

## Control

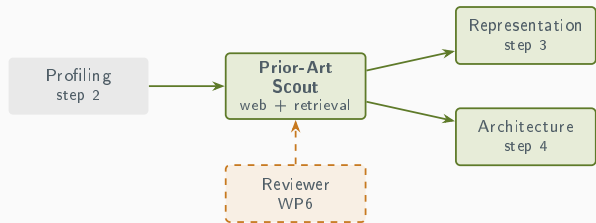
- **Reviewer:** audits after every stage; can request a retry or veto.
- **0 direct agent calls**, verified on the code's syntax tree.

# Where the LLM is allowed to decide



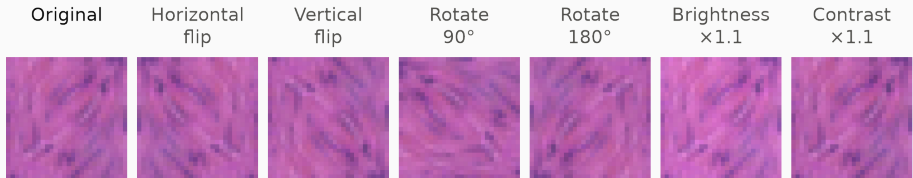
- **Deterministic:** ingestion, data audit, training, evaluation, abstention. No task-agent LLM call.
- **Judgement:** ambiguity, prior art, representation, architecture, search, report.
- Model search uses most decisions: 10 of 16.
- Every decision is logged with its prompt and source; the heuristic fallback was used once.

# The Prior-Art Scout



- Searches arXiv; every source is snapshotted and checksummed.
- Writes a cited brief that feeds representation and architecture.
- Ranks existing options, or proposes new ones that wait at a review gate.
- Advises only: never trains, selects or calls another agent.
- Live sources in the 2 Oct run: 6 of 6.

# Representation: what the model sees



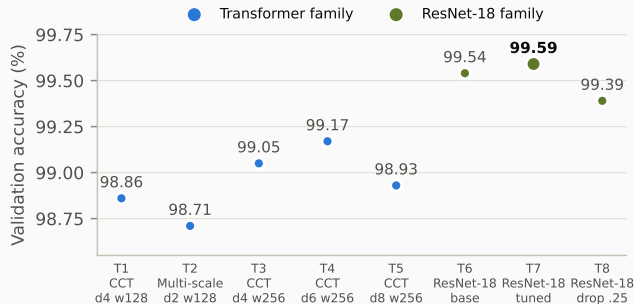
## Normalisation

- Per-channel standardisation.
- Statistics from **training images only**, reused unchanged for validation and test.

## Augmentation

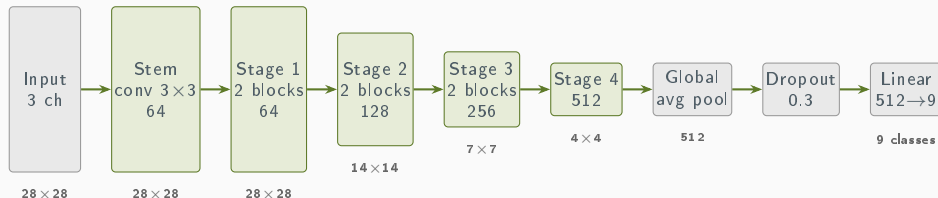
- 6 transforms picked by the representation agent from a label-preserving whitelist.
- Tissue has no fixed orientation, so flips and rotations keep the label.
- 89,996 images → 629,972 training views.

# Autonomous model search



- The agent reads every past result before choosing the next trial.
- Trials 1–5: compact and multi-scale transformers, best 99.17%.
- Trial 6: switch to ResNet-18, immediately better (99.54%).
- Selected on validation accuracy, macro-F1 as tie-break. **Test never seen.**

# The winning architecture: ResNet-18 for $28 \times 28$



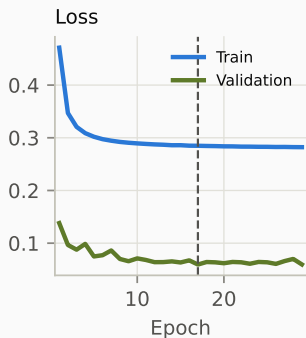
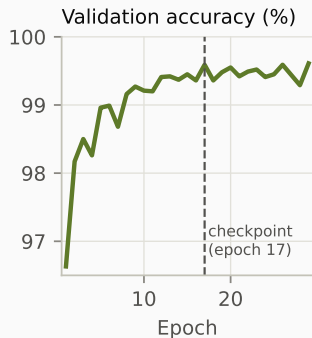
## Network

- CIFAR-style ResNet-18:  $3 \times 3$  stem and **no max-pool**, so small images keep detail.
- 4 stages  $\times$  2 residual blocks, widths  $64 \rightarrow 512$ .
- **11.17 M** parameters, trained from scratch.

## Training recipe

- AdamW, lr  $2 \times 10^{-4}$ , weight decay 0.01, cosine schedule.
- Class-weighted loss, label smoothing 0.05, dropout 0.3.
- Batch 256; early stopping with patience 12.

# Training dynamics



- Validation accuracy passes 99% by epoch 5.
- Best checkpoint: epoch 17, **99.59%**.
- Early stopping at epoch 29, after 12 epochs without improvement.
- Train loss stays near 0.28 because of label smoothing and augmentation, not underfitting.

# MVP result on the held-out hospital

| True class | ADI  | 92              |      |     |     |     | 7   |      |     |     |
|------------|------|-----------------|------|-----|-----|-----|-----|------|-----|-----|
|            | BACK |                 | 100  |     |     |     |     |      |     |     |
|            | DEB  |                 | 3    | 93  |     |     | 5   |      |     |     |
|            | LYM  |                 |      |     | 100 |     |     |      |     |     |
|            | MUC  | 4               | 2    |     |     | 92  |     | 1    |     |     |
|            | MUS  |                 | 10   | 5   |     |     | 82  |      | 2   | 1   |
|            | NORM |                 |      |     | 1   |     |     | 97   |     | 2   |
|            | STR  |                 | 1    | 13  |     |     | 17  | 2    | 52  | 14  |
|            | TUM  |                 |      |     | 2   |     |     | 2    |     | 96  |
|            |      | ADI             | BACK | DEB | LYM | MUC | MUS | NORM | STR | TUM |
|            |      | Predicted class |      |     |     |     |     |      |     |     |

**91.8%**

test accuracy

**0.981**

macro AUC

**0.890**

macro-F1

- 7,180 images from a hospital never seen in training; 95% CI 91.1–92.4%.
- 7.8 points below validation: the cost of changing hospital.
- Lymphocytes and background are perfect; cancer stroma is the weakest row.

# Leakage-safe by design



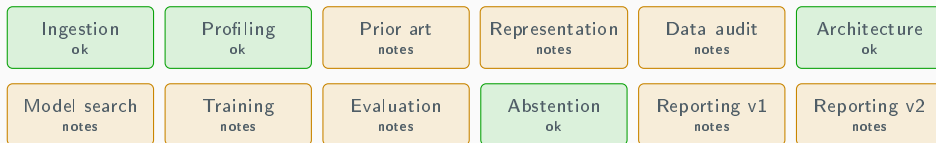
## Rules

- Normalisation fitted on training images only.
- Model selection on validation only.
- Test used once, after the model is frozen.

## Evidence in the run

- **11/11** audit checks passed before training.
- **0** images shared between train and validation.
- **0** test metrics used in selection.

# The reviewer at work



ok = no issues    notes = minor issues logged for the record

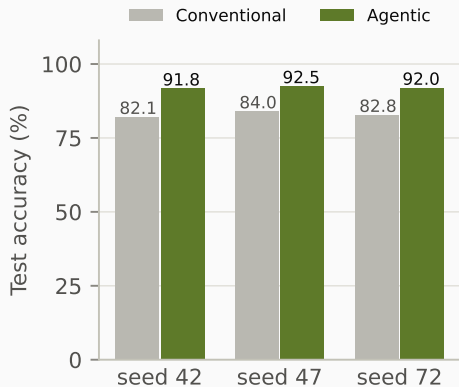
## What it checks

- Consistency and arithmetic of every artefact.
- Leakage, provenance and citation validity.
- Written claims against the actual numbers.

## What it did

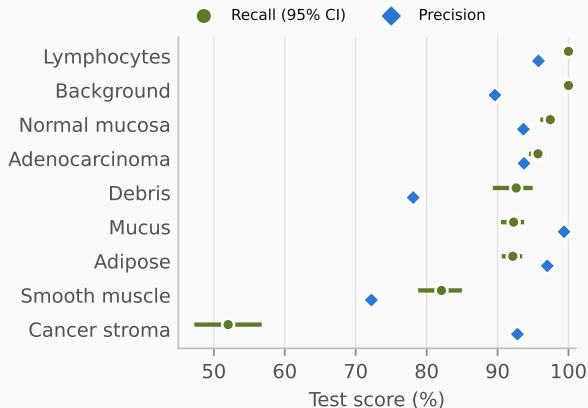
- 12 reviews: 4 clean, 8 with minor notes, **0 vetoes**.
- Caught a rationale claiming 16 image tokens instead of 49.
- Requested a rewrite of the final report (v1 → v2).

# Agents vs a conventional pipeline



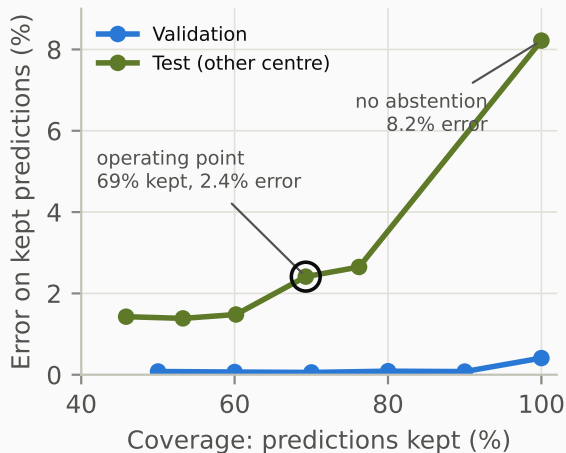
- Same data, splits and seeds; the baseline is a fixed small-CNN recipe.
- Agentic **92.1%** vs conventional 83.0%: **+9.1 points**, on every seed.
- Ablation: representation off 80.9% → standardise 86.3% → + flip 91.3%.
- Evidence: 2 Oct run (v2 pipeline), seeds 42/47/72, live arXiv sources.

# Error analysis: which tissues fail, and why



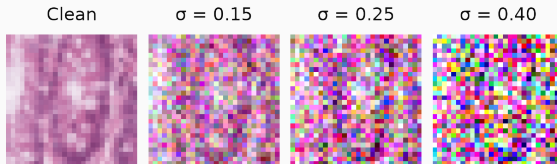
- 7 of 9 tissues reach  $\geq 92\%$  recall at the new hospital.
- **Cancer stroma**: 52% recall but 93% precision. Missed cases go to muscle (17%), adenocarcinoma (14%), debris (13%).
- At 28 pixels these tissues share fibrous texture.
- The profiling agent flagged this risk **before training**.

# Knowing when not to answer



- The model abstains when its predictive entropy is high.
- Threshold set on validation for 80% coverage; never tuned on test.
- Test: keeps 69% at **97.6%** accuracy; 2,206 images go to human review.
- Coverage drops at the new hospital: the model knows it is less sure.

# Robustness check: corrupted inputs



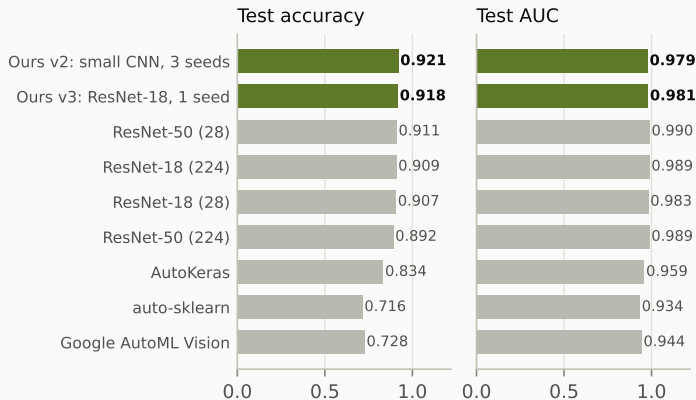
## Test

- Gaussian pixel noise at three strengths.
- Question: does the abstention score separate noisy from clean images?

## Result

- AUROC **0.943** on test (0.994 on validation).
- At most 0.2% of noisy images accepted.
- Limit: noise is a proxy; staining and scanner shifts are untested.

# Comparison with the state of the art



Published: MedMNIST v2, Yang et al., *Sci. Data* 2023, Table 3 (mean of  $\geq 3$  runs).

Accuracy: **above every published baseline.**

AUC: slightly below the best ResNets.

Far ahead of AutoML tools, the closest kind of automated pipeline.

# Acceptance verdict

| Criterion                     | 8 Oct | 2 Oct |
|-------------------------------|-------|-------|
| Typed, checksummed hand-offs  | ✓     | ✓     |
| Zero direct agent calls       | ✓     | ✓     |
| Leakage audit before training | ✓     | ✓     |
| Cited ideas                   | ✓     | ✓     |
| Live web retrieval            | –     | ✓     |
| Validation-only selection     | ✓     | ✓     |
| Baseline comparison           | –     | ✓     |
| Representation ablations      | –     | ✓     |
| Seed variance (3 seeds)       | –     | ✓     |

✓ evidenced    – not run in that mode

13 / 17  
criteria evidenced

| Criterion                      | 8 Oct | 2 Oct |
|--------------------------------|-------|-------|
| Abstention and human hand-off  | ✓     | ✓     |
| Reviewer report on every stage | ✓     | ✓     |
| Decision provenance            | ✓     | ✓     |
| End-to-end run                 | ✓     | ✓     |
| Fault detection $\geq 90\%$    | ●     | ●     |
| Full-run replay                | ●     | ●     |
| Approved risk mitigations      | ●     | ●     |
| Independent TRL 7 sign-off     | ●     | ●     |

● evidence not yet in the repository

4

need external evidence

# Thank you

Maximum cohesion, minimum coupling.